Integrating Multiple U.S. Census Bureau Data Assets to Create Standardized Profiles of Program Participants

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Abstract

The Foundations for Evidence-Based Policymaking Act of 2018 (Evidence Act) directed federal agencies to systematically use data when making policy decisions. In response, the U.S. Census Bureau established the Evidence Group within its Center for Economic Studies (CES). With an interdisciplinary team of economists, sociologists, and statisticians, the Evidence Group can support the broader federal government in their efforts to use existing data to improve program operations without increasing respondent burden. For federal agencies administering social safety net and business assistance programs in particular, the team provides a no-cost evidence-building service that links program records to Census Bureau data assets and creates a series of standardized tables describing participants, their economic outcomes prior to program entry, and the communities where they live. These tables provide partner agencies with the detailed information they need to better understand their participants and potentially make their programs more accountable and effective in reaching their target populations. In this working paper, we describe the standardized tables themselves as well as the data assets available at the Census Bureau to create these tables, the data files produced by the table production process, and the methodology used to merge and harmonize data on participants and subsequently calculate unbiased and accurate estimates. We conclude with a brief discussion of steps taken to ensure confidentiality and data security. This documentation is intended to facilitate proper use and understanding of the standardized tables by partner agencies as well as researchers who are interested in leveraging these tools to explore characteristics of their samples of interest.

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. This paper was started while Dominy was an economist at the U.S. Census Bureau and Sabelhous was a scientist with New Light Technologies. Dominy is a Senior Analyst, Economics and Industry Insights, at Cox Automotive. Sabelhous is a Data Scientist at Ibotta. The views presented in this article are those of the authors and do not represent any opinions or positions of Cox Automotive, Ibotta, or their clients. We thank Carolyn Hronis and Gloria Aldana for their contributions to the creation of the table production process and Barbara Downs and Justine Lee for their helpful comments and suggestions.

Contents

1 Background **4**

2 Summary of the table production process **4**

3 Description of the standardized tables **5**
3.1 Table set 1: Demographic profile of participants 5
3.2 Table set 2: Economic outcomes of participants prior to program entry 6
3.3 Table set 3: Socio-economic profile of communities where participants live 6

4 Input files **6**
4.1 Program data from partner agencies 6
4.2 Census Bureau data assets 9
4.2.1 Demographic data assets 9
4.2.2 Employment data assets 12
4.2.3 Geographic data assets 13

5 Data files **13**
5.1 PROG file 13
5.2 DEMO file 14
5.3 EMP file 14
5.4 GEO file 14
5.5 ACS-P file 15
5.6 ACS-H file 15
5.7 WEIGHT file 15
5.8 Note about merging data files 15

6 Methodology **16**
6.1 Determining the best value across multiple data assets 16
6.2 Creating weights to account for records missing PIKs 17
6.3 Calculating table measures 19
6.3.1 Table set 1: Demographic profile of participants 19
6.3.2 Table set 2: Economic outcomes of participants prior to program entry 19
6.3.3 Table set 3: Socio-economic profile of communities where participants live 21

7 Confidentiality protection and disclosure avoidance **24**

8 Conclusion **25**

1 Background

The Foundations for Evidence-Based Policymaking Act of 2018 (Evidence Act) directed federal agencies to systematically use data when making policy decisions. For those administering social safety net and business assistance programs, comprehensive information on participants and their outcomes can illuminate the short- and long-term impact of their programs and can be used to make them more accountable and effective in reaching their target populations.

Central to this evidence-building work is obtaining complete data on participants over time. As the nation's leading provider of quality data on its people and economy, the U.S. Census Bureau has a wealth of information on individuals and households as well as a trusted data linkage infrastructure that can securely connect outside data to internal data sources. To support the broader federal government in their efforts to use existing data to improve program operations without increasing respondent burden, the Census Bureau established the Evidence Group within its Center for Economic Studies (CES). With an interdisciplinary team of economists, sociologists, and statisticians, the Evidence Group can provide specialized data services and support to enhance program monitoring and evaluation.

To accommodate current partnerships and quickly scale future collaborations, the Evidence Group developed a production process to create a series of standardized tables that describe participants, their economic outcomes prior to program entry, and the communities where they live. These tables are created with a short turnaround time and use tested methodologies and state-of-the-art confidentiality protection. The table production process can additionally be utilized by researchers who are interested in leveraging these tools to explore characteristics of their samples of interest.

In this working paper, we describe the standardized tables themselves as well as the data assets available at the Census Bureau to create these tables, the data files produced by the table production process, and the methodology used to merge and harmonize data on participants and subsequently calculate unbiased and accurate estimates. We conclude with a brief discussion of steps taken to ensure confidentiality and data security.

2 Summary of the table production process

The table production process is designed to create three sets of standardized tables that describe participants, their economic outcomes prior to program entry, and the communities where they live. In order to do this, the process must efficiently query massive data sets on people, households, and jobs - some of which cover the entire U.S. population - and determine the most precise value for the demographic characteristics of participants. Then, it pulls earnings histories for all associated formal employment spells and identifies places of residence at the time closest to program participation. The resulting data files are then aggregated and subset to produce final tables for partner agencies. To guard

against bias due to non-random likelihoods of being matched to internal Census Bureau records, the table production process also calculates stable inverse probability weights that can be used to reweight results so they better represent all participants and not just those who were assigned a PIK. Disclosure avoidance measures are subsequently applied to all tables, which are reviewed and approved by the Disclosure Review Board (DRB) before they are released to partner agencies and the public.

3 Description of the standardized tables

This section details the three sets of standardized tables created by the table production process. All tables can be produced at the nation, region, and Census division levels.

3.1 Table set 1: Demographic profile of participants

Table set 1 provides partner agencies with a basic demographic profile of their participants and allows for comparisons between participants and a benchmark population (e.g., the general population).[2] It contains one table and shows counts and percentages of participants, overall and by the demographic characteristics listed in Table 1.

Table 1: Demographic characteristics in table sets 1 and 2

Demographic Characteristic	*Categories*	*No. of Categories*
Age[a]	5-year age groups for ages 0-84 and a pooled age group for ages 85+	18
Sex[b]	Male Female	2
Race	White alone Black or African American alone American Indian or Alaskan Native alone Asian alone Native Hawaiian or Pacific Islander alone Two or More Races	6
Ethnicity	Hispanic Not Hispanic	2

[a] Age is calculated as the difference between the participant's year of birth and the program entry year.
[b] The Decennial Census and American Community Survey (ACS) include "Some other race" as a category for race, as mandated by U.S. Congress. The standardized tables do not include this category because we use public data from the Population Estimates Program (PEP) to calculate counts and percentages for a benchmark population and PEP data does not provide counts for "Some Other Race."

[2] Table 1 relies on public data from the Population Estimates Program (PEP) to provide counts and percentages for a benchmark population. PEP counts are based on full-count Decennial Census microdata, with annual revisions for births, deaths, and migration.

3.2 Table set 2: Economic outcomes of participants prior to program entry

Table set 2 provides partner agencies with the economic outcomes of participants in the five years before program entry. It contains two tables with Table 2a showing average earnings and Table 2b showing the percent in covered employment. Participants must have at least one quarter of positive earnings from an employer covered by a state's Unemployment Insurance (UI) program in the five-year period prior to program entry to be included in these tables. This criterion excludes long-term unemployed participants so that only those with some labor market attachment are included. Participants must also be between the ages of 16 and 64 years over this entire period. This age restriction allows the tables to focus on time trends in employment and earnings that are not driven by age (e.g., aging into the labor force and retirement). Overall measures are provided, as well as breakdowns by the demographic characteristics listed in Table 1.

3.3 Table set 3: Socio-economic profile of communities where participants live

Table set 3 provides partner agencies with a socio-economic profile of the communities where participants live. It contains three tables with Tables 3a showing percent estimates for person-level categorical characteristics (e.g., educational attainment and health insurance coverage), Table 3b showing pseudo-quantile estimates for person-level continuous characteristics (e.g., personal income), and Table 3c showing percent estimates for household-level categorical characteristics (e.g., housing tenure and food stamp recipiency). All included characteristics are listed in Table 2.

4 Input files

This section outlines the input files for the table production process. We first describe program data from partner agencies, which are essential for identifying participants, the year they entered a program, and other program-specific information. We then discuss the Census Bureau data assets that provide demographic, employment, and geographic information about participants.

4.1 Program data from partner agencies

At bare minimum, program data from partner agencies must contain personally identifiable information for participants so that their records can be located in Census Bureau data assets. Due to the sensitive nature of these data, partner agencies must submit their files to a small team at the Census Bureau that operates on a highly restricted computing cluster behind a firewall. This team standardizes key linking fields like Social Security Number (SSN), name, date

Table 2: Person- and household-level characteristics in table set 3

Table	*Estimate*	*Characteristic*	*Categories*	*Universe*
3a	Percent distribution	Employment status	Employed Unemployed Armed Forces Not in labor force	Population 16 years and over
		Educational attainment	Less than high school graduate High school graduate (includes equivalency) Some college, no degree Associate's degree Bachelor's degree Graduate or professional degree	Population 25 years and over
		Health insurance coverage	Insured Uninsured	Total population
		Marital status	Now married, except separated Widowed Divorced Separated Never married	Total population
		Disability	With a disability Without a disability	Total population
		Length of commute to work	Less than 10 minutes 10 to 14 minutes 15 to 19 minutes 20 to 24 minutes 25 to 29 minutes 30 to 34 minutes 35 to 44 minutes 45 to 59 minutes 60 or more minutes	Total population

Table 2: Person- and household-level characteristics in table set 3 (continued)

Table	*Estimate*	*Characteristic*	*Categories*	*Universe*
3b	Pseudo-quantiles	Total personal income	10th percentile 25th percentile 50th percentile 75th percentile 90th percentile	Population 16 years and over
		Personal wage income	10th percentile 25th percentile 50th percentile 75th percentile 90th percentile	Population 16 years and over
3c	Percent distribution	Housing tenure	Owner-occupied Renter-occupied	Occupied housing units
		Food stamp recipiency	Receiving food stamps/SNAP Not receiving food stamps/SNAP	Occupied housing units
		Vehicles Available	No vehicles available 1 vehicle available 2 vehicles available 3 or more vehicles available	Occupied housing units

of birth, sex, and residential address and uses them to assign a unique, time-invariant identifier called a Protected Identification Key (PIK) to every record. PIK assignment is performed using probabilistic matching through the Person Identification Validation System (PVS) (Layne et al. (2014)). The SSNs and names are then removed to create a research file that is used by the table production process.

Program data from partner agencies may also include details about the programs themselves, such as name and program entry date, intensity of program utilization, and/or participant information. This allows us to curate who is included in the standardized tables and potentially produce tables focused on subpopulations of interest at certain points in time. If demographic information about participants is also included, it can be used to help reweight results so that they match distributions for all participants, rather than just the sample of participants who were assigned a PIK.

4.2 Census Bureau data assets

Using PIKs, participant records can be identified and extracted from several Census Bureau data assets spanning multiple data years (see Table 3). We primarily use three types of data assets: demographic, employment, and geographic. Demographic data sets include numerous sources, ranging from census and survey data to administrative records and composite files. Employment and geographic data rely on fewer sources and come exclusively from administrative records.

4.2.1 Demographic data assets

Program participation is often measured by looking at enrollment numbers and completion status (i.e., applied, in training, and completed). Although these statistics provide essential information for monitoring, they often lack crucial data that allow for suitable comparison benchmarks and consequently offer limited insight when evaluating programs. The Census Bureau maintains multiple census, survey, and administrative data sets with rich demographic information on nearly the entire U.S. population. By linking participant records to these existing data sources, the table production process creates a data file that can be used to generate table sets 1 and 2, which allow agencies to see participant characteristics and examine how the distributions of these characteristics differ from those of a benchmark population.

Numident The Social Security Administration (SSA) Numerical Index File, or Numident, is a master file of unique, life-long SSNs assigned to individuals based on applications for Social Security cards. It contains information on SSN applications, claim records, death information, and requested changes to SSN information. The file covers all individuals who applied for Social Security cards from the start of the program to the most recent quarter, effectively covering the entire U.S. population that has had a SSN from the inception of the program

Table 3: Data sources for the table production process

Data Type	*Data Source*	*Source Type*	*Sponsor(s)*	*Period Covered*[a]
Demographic	Numident	Administrative	Social Security Administration	All available quarters
	Decennial Census	Census	Census Bureau	2000, 2010, 2020
	American Community Survey (ACS)	Survey	Census Bureau	2005-most recent
	Current Population Survey Annual Social and Economic Supplement (CPS ASEC)	Survey	Bureau of Labor Statistics, Census Bureau	1980-most recent
	Survey of Income and Program Participation (SIPP)	Survey	Census Bureau	2014, 2018-2023
	Demographic Frame Table	Composite	Census Bureau	2015-2022
Employment	Longitudinal Employment and Household Dynamics (LEHD) - Employment History File (EHF)	Administrative	Census Bureau	1985Q1 to 3 quarters before most recent quarter
Geographic	Master Address File Extract (MAF-X)	Administrative	Census Bureau	2000-most recent
	Master Address File Auxiliary Reference File (MAF-ARF)	Administrative	Census Bureau	2000-most recent

[a] Earlier years of some data sources may be available for the table production process but are not used because they are in an unusable state or contain outdated participant records.

in 1936 through the current period. Since the Numident is cumulative, the table production process pulls in the most recent file to maximize population coverage. New files are typically available in the first month of every quarter (i.e., January, April, July, and November).

Decennial Census The Decennial Census is a census conducted in years ending in '0' to count every person living in the U.S. as of Census Day (April 1). It gathers information on family and household structure (i.e., how people live

together and are related) as well as individual demographics. The census covers every person living in the U.S., including all housing units whether occupied or vacant. The table production process pulls in Decennial Census microdata from 2000, 2010, and 2020. New microdata become available after processing.

American Community Survey (ACS) The American Community Survey (ACS) is an ongoing survey that provides vital information about our nation and its people on a yearly basis. It collects information about population and housing characteristics. The survey covers approximately 3.5 million addresses and is weighted to represent all residents of the U.S. The table production process pulls in ACS microdata from 2005 to the most recent year (which lags the current year by one to two years). New microdata are typically available in September.

ACS microdata is the only data asset used by the table production process for all three table sets. While it is one of many data sources for table sets 1 and 2, it is the primary data source for table set 3 which allows agencies to see characteristics of communities where participants live.

Best Race file The Best Race and Ethnicity Administrative Records Composite file, or Best Race file, is a composite file that combines Census Bureau surveys, administrative records, and third-party data sources. It applies business rules to assign race and ethnicity values to person records and is designed to resolve discrepancies across input data sets. It covers all individuals observed in at least one of its input data sources. The table production process pulls in the 2020 Best Race file. New files become available after processing.[3]

Current Population Survey Annual Social and Economic Supplement (CPS ASEC) The Current Population Survey (CPS) is an ongoing survey that serves as the primary source of official labor force statistics in the U.S. Its Annual Social and Economic Supplement (ASEC) is an expanded version conducted in March that provides data about the economic and social well-being of our nation and its people. The supplement covers more than 75,000 eligible households and is weighted to represent the civilian noninstitutional population 16 years and older. The table production process pulls in CPS ASEC microdata from 1980 to the most recent year (which lags the current year by one to two years). New microdata are typically available near the end of July.

Survey of Income and Program Participation (SIPP) The Survey of Income and Program Participation (SIPP) is a nationally representative longitudinal survey that provides comprehensive information on the dynamics of income, employment, household composition, and government program participation. It covers approximately 50,000 to 95,000 households and is weighted

[3]For more details on the data sources and business rules used to create the Best Race file, see Ennis et al. (2018) and Noon (2024).

to represent the civilian noninstitutional population of the U.S. The table production process pulls in SIPP microdata from 2014 to 2023. New microdata are typically available near the end of July.

Demographic Frame Tables The Demographic Frame is a comprehensive database of person-level data that contains demographic characteristics and addresses associated with individuals. It is derived from Census Bureau surveys, administrative records, and third-party data sources. It applies business rules to assign demographic values and residential identifiers to person records and is designed to resolve discrepancies across input data sets. The tables cover all individuals observed in at least one of its input data sources. The table production process pulls in Demographic Frame data from 2018 to 2022. New tables become available after processing.

4.2.2 Employment data assets

Partner agencies often ask participants about work and economic well-being to assess the impact of their programs. However, self-reported data has known limitations, with responses only capturing a single point in time and follow-up surveys frequently having low response rates (Bollinger et al. (2019)). To reliably and consistently track program participants over time, the table production process utilizes comprehensive employment information available through the Local Employment Dynamics (LED) Partnership and the Longitudinal Employment and Household Dynamics (LEHD) program.

The LED Partnership is a voluntary federal-state partnership started in 1999 to merge data about workers with data about employers to produce a collection of enhanced labor market statistics. States share UI earnings data and the Quarterly Census of Employment and Wages (QCEW) data with the Census Bureau. The LEHD program then combines these employer-reported administrative records with census, survey, and additional administrative data to create statistics on employment, earnings, and job flows at detailed levels of geography and industry and for various demographic groups.

Employment and earnings information for participants comes from LEHD's Employment History File (EHF), which contains annual records with quarterly earnings, as reported by employers, for individuals with UI-covered jobs. It covers more than 96 percent of employment in the U.S., with notable exclusions including federal government employees, independent contractors, the unincorporated self-employed, railroad workers covered by the railroad unemployment insurance system, some family employees of family-owned businesses, certain farm workers, students working for universities under certain cooperative programs, salespersons primarily paid on commission, and workers of some nonprofits. The table production process pulls in the most recent EHF file which contains data from the first quarter of 1985 to most recent quarter (which lags the current quarter by three quarters). New microdata are typically available in the first month of the quarter.[4]

[4]The LED partnership is a voluntary federal-state data sharing partnership. Data avail-

4.2.3 Geographic data assets

Master Address File (MAF) Geographic information about place of residence comes from the Master Address File (MAF), which contains an accurate, up-to-date inventory of all known living quarters in the country. An extract of the MAF called the Master Address File Extract (MAF-X) assigns a unique address identifier – MAFID – to every domicile in the U.S. and includes location latitude and longitude. The MAF-X and other data files are then used to create the Master Address File Auxiliary Reference File (MAF-ARF), which assigns one MAFID for every individual living in a domicile.[5]

We get unique MAFIDs for participants from the MAF-ARF file and latitude and longitude data for those MAFIDs from the MAF-X file. We then use TIGER/Line Shapefiles to assign year-specific Census tract identifiers to the MAFIDs. The table production process pulls in MAF-X and MAF-ARF files from 2000 to the most recent year (which lags the current year by one to two years). New MAF-X files are typically available in January and July while new MAF-ARF files are typically available between February and March, after a complete year of administrative records have been received.

5 Data files

This section provides an overview of the data files created by the table production process (see Table 4). For each file, we describe the unit of observation, explain its role in generating the standardized tables, and list the variables it contains.

5.1 PROG file

The PROG file is based on the research file and contains observations for every participant who was assigned a PIK. It is unique by participant, program, and program entry year. The purpose of this file is to provide a list of participants that will be used to determine which records should be pulled from Census Bureau data assets. It has the following variables: PIK, program, program entry year, program entry date, program exit date, and any flag variables indicating program features and/or participant characteristics that can be used to produce tables for subpopulations of interest.

ability varies by state, particularly for recent quarters. For details, see the Availability of Data section of the LEHD Snapshot Documentation.

[5]Prior to 2012, the MAF-ARF listed all known MAFIDs for an individual, meaning individuals could have multiple MAFIDs and users could decide which MAFID was the most likely address. In 2012, the MAF-ARF began listing a unique MAFID for each individual in order to meet new privacy protection and disclosure avoidance standards. Individuals with multiple MAFIDs are assigned one at random from MAFIDs that meet a minimum threshold of confidence in accuracy (Dillon (2021)).

Table 4: Data files created by the table production process

Data File	*Purpose*	*Table Set*
PROG	Contains a list of participants unique by PIK, program, and program entry year	1, 2, 3
DEMO	Contains harmonized best values for age, sex, race, and ethnicity	1, 2
EMP	Contains employer and quarterly earnings information	2
GEO	Contains geographical identifiers for place of residence	3
ACS-P	Contains ACS person-level microdata and program-adjusted person weights	3
ACS-H	Contains ACS household-level microdata and program-adjusted household weights	3
WEIGHT	Contains probability weights	1, 2

5.2 DEMO file

The DEMO file contains observations for every participant in the PROG file. It is unique by participant and program entry year. The purpose of this file is to gather demographic information about participants from multiple demographic data assets and put them into one data set for the creation of table sets 1 and 2. It has the following variables: PIK, program entry year, age, sex, race, ethnicity, nation, region, division, and state of residence.

5.3 EMP file

The EMP file contains observations for every participant in the PROG file who received at least one quarter of positive earnings from an UI-covered employer between the first quarter of 1985 and the most recent quarter (inclusive). It is unique by participant, employer, and year. The purpose of this file is to gather employer and quarterly earnings information for participants from the EHF file and put them into one data set for the creation of table set 2. It has the following variables: PIK, year, state employer identification number (SEIN), establishment identifier number (SEINUNIT), state, quarter 1 earnings, quarter 2 earnings, quarter 3 earnings, and quarter 4 earnings.

5.4 GEO file

The GEO file contains observations for every participant in the PROG file who was linked to a MAFID. It is unique by participant, program entry year, and geography level (i.e., nation, region, Census division, state, county, and Census tract). The purpose of this file is to gather Census tract information for participants from MAF files, which will be used to calculate program-adjusted person and household weights for the creation of table set 3 (see the Table set 3: Socio-economic profile of communities where participants live section).

It has the following variables: PIK, program entry year, geography level, and geographical identifier.

5.5 ACS-P file

The ACS-P file contains person-level ACS microdata. The purpose of this file is to provide the ACS microdata necessary to calculate person-level estimates for communities where participants live for table set 3, specifically Tables 3a and 3b. It also contains program-adjusted person weights, which are calculated after merging in the number of participants living in a Census tract (see the Table set 3: Socio-economic profile of communities where participants live section). The ACS-P file has the following variables: program entry year, state, county, Census tract, Census tract geographical identifier, age, sex, race, ethnicity, employment status, educational attainment, health insurance coverage, marital status, disability, travel time to work, person weight, person replicate weights, number of participants living in the Census tract, program-adjusted person weight, and program-adjusted replicate weights.

5.6 ACS-H file

The ACS-H file contains the household-level microdata. The purpose of this file is to provide the ACS microdata necessary to calculate household-level estimates for communities where participants live for table set 3, specifically for Table 3c. It also contains program-adjusted households weights, which are calculated after merging in the number of participants living in a Census tract (see the Table set 3: Socio-economic profile of communities where participants live section). The ACS-H file has the following variables: program entry year, state, county, Census tract, Census tract geographical identifier, housing tenure, food stamp recipiency, vehicles available, household weight, household replicate weights, number of participants living in the Census tract, program-adjusted household weight, and program-adjusted replicate weights.

5.7 WEIGHT file

The WEIGHT file contains an observation for every participant in the PROG file. The purpose of this file is to provide the probability that each participant was assigned a PIK given a set of demographic characteristics. This is used to correct for biases in table sets 1 and 2 caused by non-random PIK assignment (see the Creating weights to account for records missing PIKs section). The WEIGHT file has the following variables: PIK and the probability of being assigned a PIK.

5.8 Note about merging data files

The PROG, DEMO, EMP, GEO, and WEIGHT files can be merged by PIK, program, and program entry year. The GEO, ACS-P, and ACS-H files can be

merged using geographical identifiers for Census tracts. Note that the table production process does not merge the GEO file to any other data file by PIK. It is solely used to provide the number of participants living in a Census tract, which is merged into the ACS-P and ACS-H files.

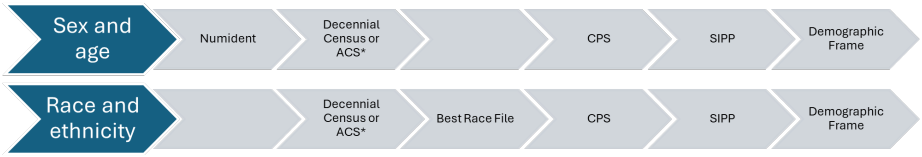
6 Methodology

6.1 Determining the best value across multiple data assets

The Census Bureau collects demographic information through numerous censuses, surveys, and administrative records. As a result, key demographic variables, such as age, sex, race, and ethnicity, often appear in several data assets. For those used by the table production process, in particular, all but the Best Race file have age and sex variables, and all have race and ethnicity variables.

Although values often agree across data assets, they do not always match. Survey responses are self-reported and individuals may respond differently between data sources and over time (Liebler et al. (2017), Fernández et al. (2018)). In addition, data sources offer different response options. For example, when asked about race, respondents can select "Some other race" for the Decennial Census and ACS but not for CPS or SIPP.

Figure 1: Hierarchical strategy to determine the "best" available source



*If both are available for the same year, use Decennial Census values over ACS.

To account for this, the table production process uses a prioritized, stepwise system to select the best available source for each participant (see Figure 1). It queries multiple data assets in a pre-determined order that varies by demographic characteristic. For race and ethnicity, it also takes program entry year into account and uses sources closest in time to actual participation. This hierarchy can easily accomodate new data assets as they become available and can be updated with a customizable set of rules (e.g., get age for participants at program entry) that consider the specific context of partner agencies. This ensures consistent, high-quality data across participants by maximizing accuracy and minimizing conflicts from multiple data assets.

If information for a participant cannot be found in any of the data assets or if the only information available is for a category not included in the standardized tables, then that variable is set to 'Missing.'

Since this strategy uses different Census Bureau data assets across multiple data years, demographic information for participants may not come from the

same data assets or data year. Given the coverage of the demographic data assets used by the table production process, most individuals will have their sex and age values sourced from the most recent Numident file and their race and ethnicity values sourced from the Decennial Census. In this case, their demographic information would come from two different data assets from two different data years. However, if an individual filled out their ethnicity but not their race in the Decennial Census, then their race value may come from the most recent ACS. Their demographic information would then come from three different data assets for up to three different data years.[6]

6.2 Creating weights to account for records missing PIKs

The Census Bureau aims to assign a PIK to every participant, but ultimately not everyone may receive one. Bond et al. (2014) note this could be "due to insufficient identifying information or because the information does not uniquely match any of the administrative records used in the person validation process." They ultimately find that PIK assignment is non-random with "young children, minorities, residents of group quarters, immigrants, recent movers, low-income individuals, and non-employed individuals [being] less likely to receive a PIK."

To correct for systematic differences in PIK assignment rates across observable characteristics, we calculate probability weights as the inverse of the probability of being assigned a PIK. These weights are then used to reweight results so that they match distributions for all participants rather than just participants who were assigned a PIK. Thus, if younger or minority participants are systematically assigned a PIK at lower rates than their counterparts, we correct for these discrepancies by assigning higher inverse-probability weights to participants who were assigned a PIK that are members of these underrepresented groups.

Before reweighting, we review the raw program data for inconsistencies indicative of measurement error (e.g., manual entry mistakes) and assess whether these issues are more prevalent among records that were not assigned a PIK. If they are, we weigh the benefits of reweighting participants who were assigned a PIK against the risk that records not assigned a PIK are prone to errors and not representative of all participants. For example, if only one to two percent of records were not assigned a PIK and inconsistencies are disproportionately concentrated within that small subset, we may forgo reweighting to avoid introducing additional measurement error.

Inverse-probability weights require us to have a reliable way of estimating the probability that a participant was assigned a PIK given observable characteris-

[6] Since the hierarchical strategy is employed for every participant for each time they participate in a program, the number of possible data assets and data years used to assign "best" values could increase if they participate in multiple programs or in the same program multiple times. For example, if an individual starts Program A in Year 1 and later joins Program B in Year 3, then the best available sources for their demographic information are determined for both Years 1 and 3. Similarly, if an individual participates in Program A in Year 1 and Year 5, then the best available sources are determined for Years 1 and 5. In both cases, up to eight different data assets for up to eight different data years could be used.

tics. To start, we first partition the research file into an 80 percent training data set and a 20 percent test data set, stratified by whether the record was assigned a PIK. Since records with an assigned PIK typically outnumber those without, we use the Synthetic Minority Oversampling Technique (SMOTE) algorithm and remove Tomek links on the training data set to address class imbalance. The SMOTE algorithm looks at records that were not assigned a PIK, finds similar records, and generates synthetic observations that interpolate between existing examples. The removal of Tomek links excludes synthetic observations that are too similar to participants who were assigned a PIK, ensuring cleaner separation between the two groups.

We then train a predictive model on the 80 percent training data set using AutoGluon, an automated ensemble of modern machine-learning models that evaluates and combines algorithms such as gradient-boosted trees, random forests, and neural networks (Erickson et al. (2020)). We then apply all trained models to the held out 20 percent test data and select the best performing model with respect to a pre-defined model fit statistic (e.g., F1 or ROC-AUC). To ensure the predictions add a meaningful signal beyond a naïve baseline, the out-of-sample F1 score on the test data set must exceed 0.60. If the top-ranked model passes this practical usefulness test, we use it to predict the probability that each participant was assigned a PIK. If not, we revert to an unweighted analysis.

While predicted probabilities from our training model may rank individuals effectively, they could systematically over- or underestimate actual likelihoods. To address this, we calibrate our predicted probabilities using the test data set. If the test data set contains at least 1,000 records and has at least 50 observations in each class, we apply isotonic regression, which adjusts probabilities while preserving their relative order. Otherwise, we use Platt sigmoid calibration, which is a statistical curve-fitting method. Both approaches align predicted probabilities with observed frequencies so that individuals with a predicted probability of 0.20 are assigned a PIK approximately 20 percent of the time.

Calibrated probabilities are also moderated to limit variance inflation and prevent extreme cases from disproportionately influencing our estimates. We first shrink predicted probabilities toward 0.5 in log-odds space (with shrinkage factor $\lambda = 0.8$), winsorize them at the 1st and 99th percentiles, and clip them to avoid exact probabilities of 1 or 0. We then compute inverse-probability weights and stabilize them by dividing each weight by their overall mean, ensuring the average weight equals one. We additionally cap the largest weights using a binary-search algorithm that systematically evaluates possible upper boundaries. This maintains stable variance levels without discarding or excessively modifying most observations.

6.3 Calculating table measures

6.3.1 Table set 1: Demographic profile of participants

Counts and percentages for table set 1 are calculated after merging the PROG, DEMO, and WEIGHT files. The universe includes all participants in the PROG file.

For demographic characteristic j, counts of participants in category k are calculated as:

$$count_{j_k} = \sum_{i=1}^{N} \mathbb{1}(j_i == k) \cdot w_i, \tag{1}$$

where:

$$N = \text{total number of non-missing observations}$$
$$w_i = \text{stable inverse-probability weight for individual } i.$$

Corresponding percentages are calculated as:

$$percent_{j_k} = \frac{count_{j_k}}{\sum_1^k count_{j_k}}.^{7} \tag{2}$$

6.3.2 Table set 2: Economic outcomes of participants prior to program entry

Measures for table set 2 are calculated after merging the PROG, DEMO, EMP, and WEIGHT files. The universe includes all participants who are observed in the EMP file at least once during the five-year period prior to program entry.[8] Repeat participants who have participated in a program multiple times are included the first time they participate in the program but are excluded in subsequent years.[9] Participants must also be between the ages of 16 and 64 years over this entire period.

Table 2a: Average earnings To calculate average earnings, we transform quarterly earnings through a multi-step process. First, since participants included in table set 2 may hold multiple jobs, we sum earnings for all jobs held by a participant in a given quarter.

Next, we adjust total quarterly earnings for inflation using the seasonally adjusted Consumer Price Index for All Urban Consumers (CPI-U) series from the Bureau of Labor Statistics.[10] Since the CPI-U is a monthly series, the CPI-U is averaged over the three months in each quarter to create a quarterly CPI-U

[7] Percentages are calculated using noisy counts, as required by the DRB's policy on use of 2020 Decennial Census data.

[8] This restriction creates a selected sample of individuals with at least some UI-covered employment before program entry. As a result, those with no prior UI-covered employment history are excluded from table set 2.

[9] This is done to keep the selected sample clean from any possible treatment spells.

[10] As of this writing, the base quarter is 2024Q1.

measure. For a given quarter q and base quarter $qbase$, total quarterly earnings are adjusted as follows:

$$total\ earn^{q}_{real} = \frac{total\ earn^{q}_{nominal}}{[\frac{cpi_q}{cpi_{qbase}}]}. \tag{3}$$

Then, we create an event-time index that indicates the year prior to program entry (i.e., year -5, year -4, year -3, year -2, and year -1). Years are indexed to the quarter when participants entered the program – the start reference quarter – and include the four quarters preceding the start reference quarter for a given year. Thus, if a participant participated in a program from September 2018 to August 2019, the starting reference quarter would be Q3, the first year in the table (i.e., year -5) would range from 2013Q3-2014Q2, and the last year (i.e., year -1) would range from 2017Q3-2018Q2.

Last, we annualize earnings by summing the four CPI-adjusted total quarterly earnings for a given event-time index.

For demographic characteristic j, average earnings in event-time t for participants in category k are calculated as:

$$\overline{annual\ earn}_{j_k t} = \frac{\sum_{i=1}^{N_t} \mathbb{1}(j_i == k) \cdot annual\ earn_{it} \cdot w_i}{\sum_{i=1}^{N_t} \mathbb{1}(j_i == k) \cdot w_i}, \tag{4}$$

where:

$$N_t = \text{total number of non-missing observations in event-time } t$$
$$w_i = \text{stable inverse-probability weight for individual } i.$$

Table 2b: Percent in covered employment For demographic characteristic j, the percent in covered employment in event-time t for participants in category k is calculated as:

$$\%\ employed_{j_k t} = \frac{\sum_{i=1}^{N_t} \mathbb{1}(j_i == k) \cdot LFP_{it} \cdot w_i}{\sum_{i=1}^{N} \mathbb{1}(j_i == k) \cdot w_i},^{11} \tag{5}$$

where:

$$N_t = \text{total number of non-missing observations in event-time } t$$
$$LFP_{it} = \begin{cases} 1, & \text{if } annualearn_{it} > 0 \\ 0, & \text{otherwise} \end{cases}$$
$$w_i = \text{stable inverse-probability weight for individual } i.$$

[11] Percent in covered employment is calculated using noisy counts, as required by the DRB's policy on use of 2020 Decennial Census data.

6.3.3 Table set 3: Socio-economic profile of communities where participants live

Whereas table sets 1 and 2 describe participants, table set 3 focuses on the communities where they live. To calculate these estimates, we leverage existing ACS survey design and scale ACS person and household weights to reflect the actual distribution of where participants live. As a result, we are able to accurately portray the characteristics of participant residences - closest in time to program participation - and give more weight to places where they are clustered.

Program-adjusted weights for each Census tract are constructed to maintain the following three properties:

1. Program-adjusted weights equal zero for all individuals in Census tracts with no participants.
2. When participants are present in a Census tract, program-adjusted weights for all individuals in that tract remain proportional to their original values.
3. For each Census tract, the sum of program-adjusted weights for all individuals in that tract are proportional to the number of participants in that tract.

In line with the second property, ACS person weights are adjusted using a constant factor, such that:

$$W_{ig}^{adj} = W_{ig} \cdot f_g, \tag{6}$$

where:

$$\begin{aligned} W_{ig}^{adj} &= \text{program-adjusted person weight of individual } i \\ W_{ig} &= \text{ACS person weight of individual } i \\ f_g &= \text{constant factor for Census tract } g. \end{aligned}$$

This can be aggregated for Census tract g such that:

$$\sum_{i=1}^{N_g} W_{ig}^{adj} = \sum_{i=1}^{N_g} W_{ig} \cdot f_g, \tag{7}$$

where

$$N_g = \text{total number of participants in tract } g.$$

In line with the third property, the right-hand side can be rewritten such that:

$$\sum_{i=1}^{N_g} W_{ig} \cdot f_g = p_g \cdot c, \tag{8}$$

where:

$$\begin{aligned} p_g &= \text{total number of participants in Census tract } g \\ c &= \text{arbitrary constant.} \end{aligned}$$

Setting c to 1, we define the constant factor for Census tract g as:

$$f_g = \frac{p_g}{\sum_{i=1}^{N_g} W_{ig}}. \tag{9}$$

Constant factors for household weights and all replicate weights are defined similarly.

The total number of participants in every Census tract observed is calculated using the GEO file. This is then merged into the ACS-P and ACS-H files to calculate program-adjusted person and household weights. Measures for Tables 3a and 3b are then calculated using the ACS-P file while measures for Table 3c are calculated using the ACS-H file.

Table 3a: Percent estimates of person-level categorical variables Table 3a displays percent distributions, which are calculated by first determining estimates and then computing percent estimates.[12]

For characteristic j, estimates of individuals in category k residing in Census tracts where participants live are calculated as:

$$estimate_{j_k}^{adj} = \sum_{i=1}^{N} \mathbb{1}(j_i == k) \cdot W_i^{adj}, \tag{10}$$

N = total number of individuals with non-missing observations

W_i^{adj} = program-adjusted person weight for individual i.

Corresponding percentages are calculated as:

$$percent_{j_k} = \frac{estimate_{j_k}}{estimate_j}. \tag{11}$$

Corresponding margin of errors (MOEs) are calculated at the 90 percent confidence level as:

$$MOE_{j_k} = \pm 1.645 \cdot \sqrt{\frac{4}{80} \sum_{r=1}^{80} (percent_{j_k}^r - percent_{j_k})^2}, \tag{12}$$

where $percent_{j_k}^r$ is the percentage calculated using the program-adjusted replicate weight r instead of the program-adjusted person weight.[13]

[12] Percentages for table set 3 are calculated using unrounded, weighted estimates.

[13] A margin of error is a measure of an estimate's variability. The larger the margin of error in relation to the size of the estimate, the less reliable the estimate. When added to and subtracted from the estimate, the margin of error forms the 90 percent confidence interval. ACS margin of errors are calculated for table set 3 using the Successive Differences Replicate Methodology (SDR), which has the advantage that the variance estimate may be computed without consideration of the form of the statistic (e.g., count, mean, ratio, etc.) or the complexity of the sampling and weighting procedures. It relies on the ACS estimate and 80 replicate estimates. For an overview on the SDR formula, see "Calculating Margins of Error the ACS Way" webinar.

Table 3b: Pseudo-quantile estimates of person-level continuous variables Table 3b displays pseudo-quantile estimates, which are calculated by first sorting the ACS-P file in ascending order. We then identify the observation at qth percentile using the weighted distribution. Next, we calculate a weighted average of this observation and the five observations directly above and below it (using the unweighted distribution).

For characteristic j in Table 3b, the pseudo-quantile estimate is calculated as:

$$\tilde{Q}_j = \frac{1}{11} \cdot \sum_{i=i^*-5}^{i^*+5} Y_{ij} \cdot W_i^{adj}, \tag{13}$$

where:

$$\begin{aligned} i^* &= \text{observation index at the } q^{th} \text{ percentile} \\ Y_{ij} &= \text{value for characteristic } j \text{ for individual } i \\ W_i^{adj} &= \text{program-adjusted person weight for individual } i. \end{aligned}$$

Corresponding MOEs are calculated at the 90 percent confidence level as:

$$MOE_j = \pm 1.645 \cdot \sqrt{\frac{4}{80} \sum_{r=1}^{80} (\tilde{Q}_j^r - \tilde{Q}_j)^2}, \tag{14}$$

where $\tilde{Q}_j^r$ is the pseudo-quantile calculated using the program-adjusted replicate weight r instead of the program-adjusted person weight.

Table 3c: Percent estimates of household-level categorical variables Table 3c displays percent distributions, which are calculated by first determining estimates and then computing percent estimates.

For characteristic j, estimates of households in category k residing in Census tracts where participants live are calculated as:

$$estimate_{j_k}^{adj} = \sum_{i=1}^{N} \mathbb{1}(j_h == k) \cdot W_h^{adj}, \tag{15}$$

where:

$$\begin{aligned} N &= \text{total number of households with non-missing observations} \\ W_h^{adj} &= \text{program-adjusted household weight for household } h. \end{aligned}$$

Corresponding percentages are calculated as:

$$percent_{j_k} = \frac{estimate_{j_k}}{estimate_j}. \tag{16}$$

Corresponding MOEs are calculated at the 90 percent confidence level as:

$$MOE_{j_k} = \pm 1.645 \cdot \sqrt{\frac{4}{80}\sum_{r=1}^{80}(percent^r_{j_k} - percent_{j_k})^2}, \tag{17}$$

where $percent^r_{j_k}$ is the percentage calculated using program-adjusted replicate weight r instead of the program-adjusted household weight.

7 Confidentiality protection and disclosure avoidance

To protect the confidentiality of participants, standardized tables are reviewed by the DRB and must meet all standards established by disclosure avoidance policies before they are approved for public release.

The following disclosure avoidance measures are implemented by the table production process:

- Counts in Table 1 and the "Overall" rows in table set 2 are noise-injected using the Discrete Gaussian Mechanism (DGM), a DRB-approved noise injection mechanism for releasing counts and ratios of counts.[14] All noisy counts have a lower bound of 0.
- Percentages in Table 1 and the percent in covered employment in Table 2b are based on counts that have been noise-injected using DGM. All percentages based on noisy counts have a lower bound of 0 and an upper bound of 100.
- Average earnings in Table 2a and the percent in covered employment in Table 2b are not provided if they are based on less than 50 observations.
- Table set 3 relies on ACS microdata and consequently shows weighted estimates and their margins of error (MOEs).
  - Weighted percent estimates and MOEs in Tables 3a and 3c are suppressed if they are based on one or two observations.
  - Pseudo-quantile estimates and MOEs in Table 3b are based on 11 observations (the true quantile observation and the five observations above and below it) rather than the true quantile observation alone.
  - All MOEs that round to 0.0 in Tables 3a and 3c or 0 in Table 3b are replaced with values of 'Z' so users are aware that they are not true zeros.

[14] Default parameters for the DGM are set according to the recommendation of the Data Stewardship Executive Policy (DSEP) Committee or its delegate, with the DRB having authority to require that certain outputs use parameters that result in a higher variance than implied by the recommendation.

- Average earnings in Table 2a and pseudo-quantile estimates in Table 3b are rounded to four significant digits. Percentages in Table 1, the percent in covered employment in Table 2b, and percent estimates in Tables 3a and 3c are rounded to the tenth decimal place.

8 Conclusion

The table production process presented in this working paper demonstrates a robust and scalable approach to integrating multiple Census Bureau data assets to create high-quality, standardized tables about participants, their economic outcomes prior to program entry, and the communities where they live. Its increased use by federal agencies will not only promote the use of existing data to improve program operations without increasing respondent burden but also ensure they have the detailed information they need to better understand their participants and potentially make their programs more accountable and effective in reaching their target populations.

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